MARCELL MOLNÁR

Quantitative Research · Model Validation · Data Science & Statistical Learning

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Quantitative researcher and model validator working across probability, statistics, machine learning and time-series modelling. Over a year of hands-on model risk experience in Hungarian banking (Raiffeisen, OTP), an MSc in economic and financial mathematics and a parallel mathematics BSc, plus a portfolio of reproducible research from limit-order-book forecasting to Bayesian and survival models.

PROFESSIONAL EXPERIENCE

Risk Model Validator, Full Time · **Raiffeisen Bank** *July 2026 – Present*

- Independently review statistical and machine-learning models end to end — methodology, assumptions, performance, implementation, controls and documentation.
- Challenge model design and validation evidence, quantify limitations, and present findings to stakeholders in a model-risk and regulatory (MNB/EBA) context.

Risk Model Validator Intern · **OTP Group** *Jun 2025 – May 2026*

- Validated credit, market and liquidity risk models using statistical, econometric and time-series techniques.
- Built configurable Python validation tooling and co-developed group-level standards for non-credit model validation and development.

Teaching Assistant · **Corvinus University of Budapest** *Feb 2024 – Jun 2025*

- Taught and supported seminars in econometrics, time-series analysis and statistics.

EDUCATION

MSc, Economic & Financial Mathematical Analysis · **Corvinus University of Budapest** *2023 – 2028 (exp.)*

- Quantitative finance, probability, statistics, econometrics, time series, deep learning, algorithm theory.

BSc, Mathematics · **University of Szeged — Bolyai Institute** *2026 – 2028 (exp.)*

- Parallel degree taken alongside the MSc, focused on rigorous mathematical foundations.

SELECTED PROJECTS

- **Regime LOB** Market microstructure / ML — leakage-aware limit-order-book forecasting with purged temporal splits, causal regime discovery and a temporal convolutional network.
- **Symbolic Survival Credit Risk** Risk modelling — auditable discrete-time default-hazard models with temporal validation, calibration diagnostics and reproducible public data.
- **ShiftLab** Online learning — prequential benchmark of adaptive forgetting under regime shift; implemented twice (Python, OCaml) agreeing to 2×10^{-11}.
- **Bayesian Election Modelling** Bayesian statistics — hierarchical Dirichlet MCMC framework for polling uncertainty and election dynamics.
- **EU Regional Educational Inequality** Spatial econometrics / ML — spatial panel modelling, autocorrelation and convergence analysis across European regions.
- **Budapest Markov Passenger Flow** Stochastic processes / networks — GTFS-derived continuous-time Markov chains, flow simulation, centrality and network resilience under node removal.
- **LatencyEdge** ML systems — reproducible benchmark of teacher–student distillation and dynamic int8 inference, mapping accuracy–latency trade-offs on CPU.

ONGOING RESEARCH

- Manuscripts in development: hierarchical Dirichlet–multinomial spline modelling of Hungarian election polls · nonparametric model diagnostics via multifidelity Kolmogorov–Arnold networks (permutation-based conditional bias testing) · contraction-hierarchy time-series trading strategy · Bayesian modelling of US elections.

TECHNICAL SKILLS

- **Programming** Python (pandas, NumPy, PyTorch, scikit-learn, statsmodels) · R · SQL · C++ · VBA · OCaml
- **Quantitative** probability · statistics · time series · econometrics · Bayesian inference & MCMC · Monte Carlo · survival analysis · spatial statistics · model validation
- **ML & Tools** deep learning · temporal modelling · symbolic regression · online learning · Git · Databricks · Stan · LaTeX · Tableau · Jira/Confluence

AWARDS, LEADERSHIP & LANGUAGES

- **2nd place**, GEM × Morgan Stanley Statistics Competition (2025) · IMC Prosperity and Everesteer Hackathon participant.
- **Vice President of Finance & Corporate Relations**, FAKT College for Advanced Studies — led budgeting and financial planning for a 10–15 person division.
- **Languages** Hungarian (native) · English (C1 Advanced) · German (B2 Upper-Intermediate)